

LOW FAT, LOW CHOLESTEROL DIET IN SECONDARY PREVENTION
OF CORONARY HEART DISEASE

J.M. Woodhill, A.J. Palmer, B. Leelarthapin,
C. McGilchrist and R.B. Blacket

Department of Medicine, University of New South
Wales, Prince Henry Hospital, Little Bay,
Australia

SUMMARY

Four hundred fifty-eight men with coronary heart disease participated in a trial of secondary prevention for 2 to 7 years. Overall five year survival was 81%. For those with first heart attacks it was 86%.

From infarct to entry into the trial the majority of men had already made changes in their diet and smoking habit and has lost weight. They were allocated randomly to two dietary groups. In one group the diet consumed derived 9.8% of calories from saturated fatty acids and 15.1% from polyunsaturates. In the second group saturated fatty acids contributed 13.5% and polyunsaturated fatty acids 8.9% of total calories. Survival was slightly better in the second group. Multivariate analysis showed that none of the dietary factors were significantly related to survival. Prognosis was determined largely by the extent of the coronary and myocardial disease as judged by the usual clinical parameters. Recreational physical activity had a strong favourable influence on survival when all other factors were kept constant.

Although body weight and cigarette smoking were not significantly related to survival there are grounds for the belief that relative leanness and low cigarette consumption may have had a favourable influence in both dietary groups.

It is concluded that because of multiple changes in lifestyle men who have had myocardial infarction are not a good choice for testing the lipid hypothesis. Weight loss, reduction in cigarette smoking, increase in physical activity and other readjustments may well have more important beneficial effects than change in dietary lipids.

The Sydney Diet-Heart Study was conceived in 1964. At that time there was a strong belief among epidemiologists and medical scientists that the high dietary intake of saturated fats and cholesterol in the Western nations was a major determinant of premature vascular disease, especially angina pectoris and myocardial infarction. It was thought that differences in diet might explain, at least in part, the five- to ten-fold differences in reported coronary mortality between populations with a Northern European culture and those in the Mediterranean region and in Asia (1). It has also been established that substitution of polyunsaturated fatty acids for saturated fatty acids in the diet lowered serum cholesterol. The American Heart Association had recommended changes in the American diet appropriate to these beliefs. Data from a variety of sources also implicated cigarette smoking and high blood pressure as important risk factors.

The dietary hypothesis had never been submitted to a field test. There was much debate then and now on its relevance to the cardiac problems of Western man. Obesity was given a low rating as a risk factor despite the obvious adiposity of Western man and statistics from life insurance companies showing its unfavourable effects on longevity (2). It was apparent that the lifestyle of Western man must bear on the etiology of vascular disease. The complexity of the risk factors, known and unknown, and their interactions in people with different genetic make-up was only dimly appreciated.

Feasibility studies in our department had shown that it was possible to modify the diet in properly motivated people along the lines suggested by the American Heart Association. It seemed that application of the new dietary principles to men who had already demonstrated their proclivity to premature coronary disease might so improve their survival that the dietary hypothesis would gain substantial support.

The Sydney Diet-Heart Study was begun in 1966. Four hundred fifty-eight men aged 30-59 with clinical coronary disease were studied for two to seven years. They were accepted into the study not less than eight weeks (mean eleven weeks) after their most recent coronary episode. Three hundred ninety-five men (86%) had experienced myocardial infarction as judged by a typical history, ECG and enzyme changes. Sixty-three men (14%) has recent-onset angina or had experienced acute coronary insufficiency with ECG but without enzyme changes. Two hundred seventy-six (60%) of the 458 men had had no previous cardiac symptoms. One hundred eight-two (40%) had had probable or confirmed angina or myocardial infarction previously.

After work-up the men were allocated by random numbers to one of two dietary regimens. Two hundred thirty-seven men (the P group) were given no specific dietary instruction apart from restriction of calories if thought to be overweight. As a concession to the popular and medical beliefs of the time they were allowed to use polyunsaturated margarine instead of butter if they wished. Two hundred twenty-one men (the F group) were advised and tutored individually to reduce saturated fat intake to approximately 10% of calories and dietary cholesterol to 300 mg or less per day. They were encouraged to use food containing polyunsaturated fatty acids to 15% or more of daily calories. The diets of all participants were assessed by interview and/or food log three times during the first year and twice yearly thereafter. The dietary data were converted to mean daily nutrient intake on a previously established computer programme and utilized in the subsequent analyses of survival.

Table 1 shows some characteristics of the two dietary groups on entry and during follow-up. They were well-matched for age, infarct weight, entry weight, systolic and diastolic blood pressure, serum cholesterol, serum triglyceride and serum urate. They were also well-matched for disability index, cigarette smoking, previous coronary disease, occupation and physical activity. There was no difference in entry characteristics, nor in survival between men whose most recent episode was myocardial infarction and men in whom it was angina or acute coronary insufficiency. The latter have been included with the former in all analyses. There had been appreciable and significant loss of weight in both groups from infarct to entry and further loss occurred during follow-up. Significant changes also occurred in smoking habit. These are discussed later.

Table 1

SOME CHARACTERISTICS OF THE TWO DIETARY GROUPS, P AND F
AT ENTRY AND DURING FOLLOW-UP. MEANS $\pm$ S.D.*

	Group P	Group F
	<u>N=237</u>	<u>N=221</u>
<u>ENTRY</u>		
Age	49.1 $\pm$ 6.5	48.7 $\pm$ 6.8
Weight at infarct	75.9 $\pm$ 11.4	76.4 $\pm$ 10.6
Weight at entry	73.2 $\pm$ 9.5	73.1 $\pm$ 9.3
Systolic blood pressure	136.9 $\pm$ 21.1	136.6 $\pm$ 20.1
Diastolic blood pressure	88.5 $\pm$ 12.0	88.5 $\pm$ 12.5
Serum cholesterol	282.0 $\pm$ 55.6	281.3 $\pm$ 63.4
Serum triglyceride	185.9 $\pm$ 132.9	189.0 $\pm$ 199.1
Serum urate	6.7 $\pm$ 1.6	6.7 $\pm$ 1.6
<u>FOLLOW-UP</u>		
Mean weight	71.3 $\pm$ 8.5	71.4 $\pm$ 8.6
Systolic blood pressure	137.9 $\pm$ 16.1	136.8 $\pm$ 16.1
Diastolic blood pressure	88.7 $\pm$ 9.6	88.4 $\pm$ 9.4
Serum cholesterol	262.3 $\pm$ 42.0	250.2 $\pm$ 44.7**
Serum triglyceride	154.5 $\pm$ 76.5	144.4 $\pm$ 80.9

* During follow-up mean weight, serum cholesterol and serum triglyceride were significantly lower in both groups than on entry.

** $p > 0.01$.

Sixty-seven died during follow-up. Sixty deaths were due to coronary disease, three to cerebral vascular disease, two to cancer and two to motor accidents. In one of the latter death was thought due to sudden cardiac death. Twenty-eight deaths occurred in the P group and 39 in the F group. Overall 5 year survival of the 458 men was 81% (Figure 1). For the 276 men without hint of previous cardiac symptoms five year survival was 86% giving an attrition rate of approximately 3% per annum. Cumulative percentage survival of the P and F groups is shown in Figure 2. Survival was significantly better in the P group, although the difference was not marked.

Because of the apparent lack of influence of diet on

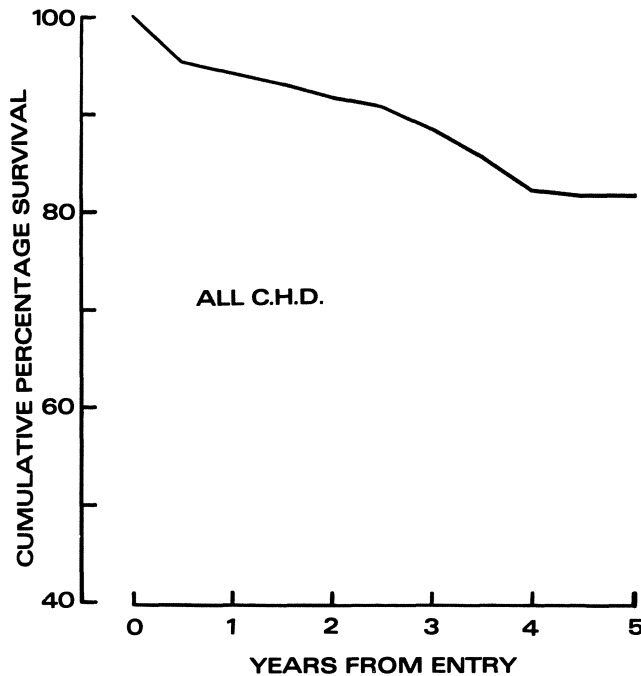


Figure 1: Cumulative percentage survival of 458 men with coronary disease followed over 5 years.

survival univariate analysis of the data of the whole group was done to determine the factors present on entry which might have prognostic significance. The following were unfavourable for survival when each was considered as a single variable; previous coronary disease, severe as opposed to mild or moderate infarction, presence at entry of angina pectoris or unusual dyspnoea on exertion, cardiac enlargement, ST-T and Q wave abnormalities in the ECG, excessive weight, hyperuricemia. Survival according to severity of angina is shown in Figure 3. Age, systolic blood pressure, serum cholesterol, serum triglycerides, glucose tolerance, smoking habit, history of premature coronary disease in first degree relatives and site of infarction by ECG were not related to survival. It was clear that the extent of the coronary disease and the functional performance of the myocardium were the major determinants of the prognosis.

The diets being consumed on entry to the study by

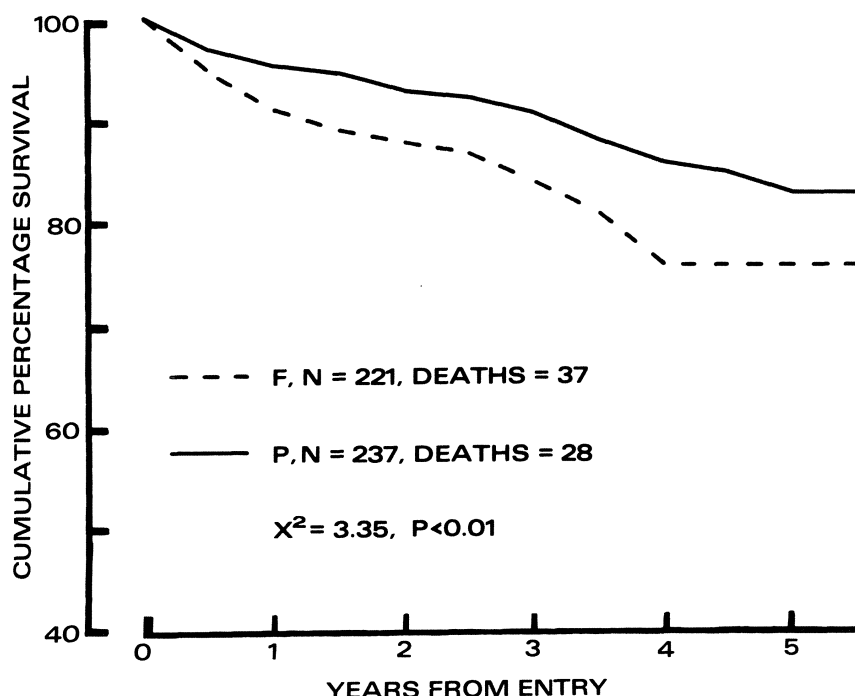


Figure 2: Cumulative percentage survival of the dietary groups P and F during 5 years of follow-up.

the P and F groups are shown in Table 2. The pattern of the nutrients conformed well to that found in normal subjects in Australia (3). However, there were slight but probably important differences. The caloric intake was lower, the saturated fatty acids and cholesterol intake were lower and the polyunsaturated fatty acid intake was higher than in normal healthy men. These changes had been made following infarction and before attending our clinic. There were no significant differences between the diets of the two groups on entry. Our estimate was that mean intake had declined by 500 calories or more per day.

Change in diet after infarct was reflected in body weight on entry. Three hundred seven of the 458 men had

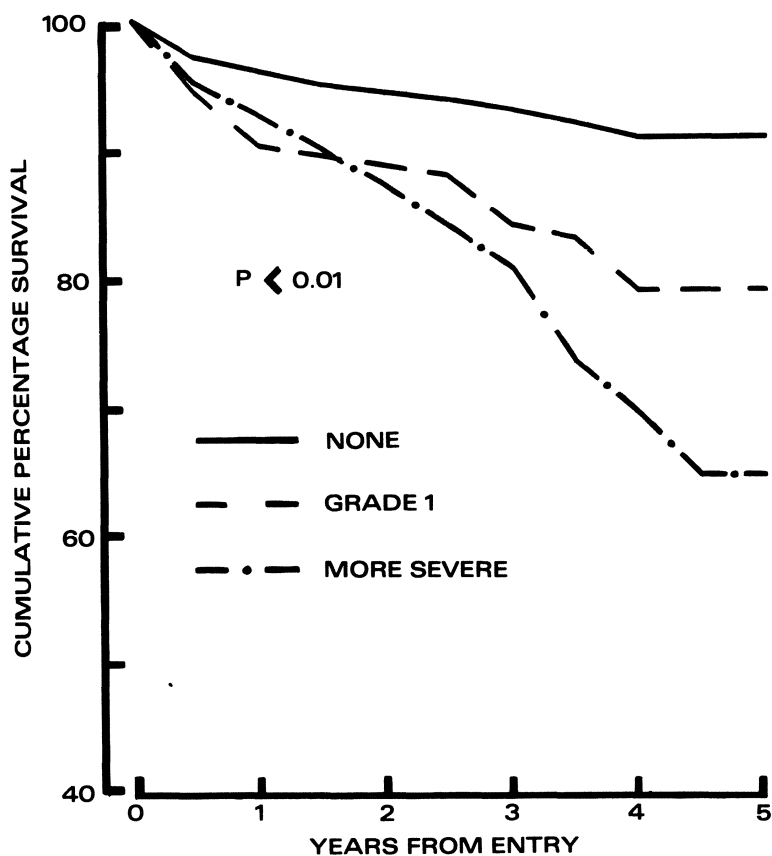


Figure 3: Cumulative percentage survival by grade of angina pectoris on entry.

Grade 1 = angina while hurrying on the level or climbing stairs or hills at normal speed.

Table 2

MEAN COMPOSITION OF THE DIET IN EACH GROUP ON ENTRY*

	GROUP P N=221		GROUP F N=205	
	% CALS.	GRAMS	% CALS.	GRAMS
Protein	14.7±3.8	85±21	14.7±3.0	87±19
Fat	39.4±6.0	105±30	39.9±6.6	108±31
Sats.	15.9±4.2	43±16	16.4±4.1	45±16
Monos	14.9±3.1	40±12	14.9±2.9	40±11
PUFA	6.6±3.8	18±11	6.6±4.1	18±13
Carbohydrate	40.8±6.9	246±75	40.3±8.3	253±101
Alcohol	5.0±6.1	18±25	5.0±6.4	18±26
Cholesterol (mg)		486±214		508±226
Calories		2397±602		2457±645
P/S		0.5±0.4		0.5±0.4

* None of the differences were significant. The data from 16 men in the P group and 16 men in the F group were considered unreliable and have been omitted.

lost a mean of 5.4 kg since infarction; they lost a further 1.4 kg during follow-up. One hundred fifty-one men had gained a mean of 2.3 kg after infarction; they lost a mean of 2.5 kg during follow-up. Both the P and F groups were lighter than they had been at infarction and had already made changes in their diet prior to randomization.

Table 3 shows the dietary patterns of the two groups during follow-up. There was a fall in mean daily intake of approximately 200 calories per day in both groups. The F group showed changes in saturated fat, polyunsaturated oils and dietary cholesterol which, on average, agreed well with the dietary prescription. Unfortunately the P group, thought of as controls, also changed in the same

Table 3

MEAN COMPOSITION OF THE DIET IN EACH GROUP ON FOLLOW-UP^a

	GROUP P N=221		GROUP F N=205	
	% CALS.	GRAMS	% CALS.	GRAMS
Protein	15.7±3.4		15.2±2.8	
Fat	38.1±5.4	83±17	38.3±5.9	85±16
Sats.	13.5±3.2	92±23	9.8±2.6 ^c	96±22 ^b
Monos	13.8±2.5	33±11	11.5±2.1 ^c	25±8 ^c
PUFA	8.9±3.5	33±9	15.1±4.3 ^c	29±7 ^c
Carbohydrate	40.3±7.3	22±10	40.9±7.3	38±13 ^c
Alcohol	5.7±5.8	226±81	5.4±5.8	240±83
Cholesterol (mg)		19±21		18±20
Calories		342±103		248±75 ^c
P/S		2191±549		2289±508
		0.8±0.5		1.7±0.6 ^c

^a Both the percentage composition and the absolute amount of the nutrients were significantly different from those on entry in both groups for nearly all the nutrients at the 0.001 level. Alcohol and percentage carbohydrate were not significantly different.

^b $p < 0.05$

^c $p < 0.001$

direction but to a lesser extent. Weight loss was 1.9 kg in the P group and 1.7 kg in the F group. The P group was now 4.6 kg and the F group 4.8 kg lighter than at infarction. Serum cholesterol declined by 19.7 mg/100 ml in the P group and by 31.1 mg/100 ml in the F group (Table 1). There was no significant change in blood pressure from entry to follow-up in either group.

Table 4 shows the profound change that occurred in

Table 4
SMOKING HABIT AT INFARCTION AND ON ENTRY
TO THE DIET-HEART STUDY

	AT INFARCTION	ON ENTRY
Non-smokers	137 (30%)	284 (62%)
Smokers	321 (70%)	174 (38%)
Total	458	458

smoking habit. Non-smokers increased from 30% at infarction to 62% on entry. Those who continued to smoke reduced their intake drastically. At infarction 47.1% of the total group were smoking more than 20 cigarettes a day. At entry only 6.1% exceeded that usage.

It is apparent that the difference in diet between P and F groups was smaller than we had hoped. It was unlikely to show significant advantage for the F group in the numbers studied. However there was considerable intra group variation in diet and overlap between the groups. To determine whether those with the higher fat diets did worse than those with a low fat diet multivariate analysis by the method of stepwise regression in life tables was done (4). In this method the diets of subjects who die are compared with the diets of those living at the time of each death. No significant differences were found. Simple comparison of the mean diet of those who died with those who survived also revealed only trivial differences. There was a trend towards better survival of those with the lowest levels of serum cholesterol but this did not quite reach the 5% level of significance. Serum cholesterol level was significant for survival in the control subjects of the Coronary Drug Project (5). However it was low on the list of determinants and very large numbers were needed to show its influence.

Clearly what was planned as a test of the lipid hypothesis became a multifactorial study. Changes in smoking habit, dietary pattern, body weight, lifestyle and physical activity before and after entry to the trial may well have had a significant effect on prognosis. On that background any added effect of further dietary change could

not be shown. Both the number of subjects and the duration of the study may have been insufficient. However, the force of strict fat modification must be low if very large numbers are needed to show its benefit in secondary prevention. More drastic restriction of fat and cholesterol to the levels prevailing in Asian countries and elsewhere might yield a positive answer. General acceptance of such change is unlikely.

In an effort to determine the separate influences of the large number of variables, medical, behavioural and dietetic data from infarction to death or the conclusion of the study were submitted to a number of multivariate analyses using the same method of stepwise regression in life tables (4).

Preliminary computer analyses were directed to reduction of the variables to manageable numbers so that only those which had any likelihood of being important were included in the final run. In addition a number of entry variables of interest were included. The final input comprised 26 entry variables plus 10 medical variables and 12 dietary variables during follow-up. By this method of analysis the entry variables which were associated with poor survival were, in order, the use of digitalis, the presence of angina, marital separation or divorce, history of diabetes, Q and ST-T changes in the ECG, left ventricular hypertrophy by ECG and the presence of clinical cerebrovascular disease. Other indicators of myocardial disease found significant by univariate analysis, such as severity of infarction and radiological cardiac enlargement, were accounted for by these variables. The factors which were non-contributory on univariate analysis remained non-contributory on multivariate analysis.

During follow-up serum cholesterol, serum triglycerides, body mass index, blood pressure and smoking habit had no predictive value. Recreational physical activity had a highly favourable effect. The use of diuretics for the first time during follow-up had an unfavourable effect. The development of new angina pectoris, the new use of digitalis and the appearance of any degree of cardiac failure were also unfavourable but added little to the prediction of probability of death. As found previously none of the dietary variables had predictive power. The multivariate analysis confirmed the dominant effect on prognosis of the coronary and myocardial disease present on entry to the study.

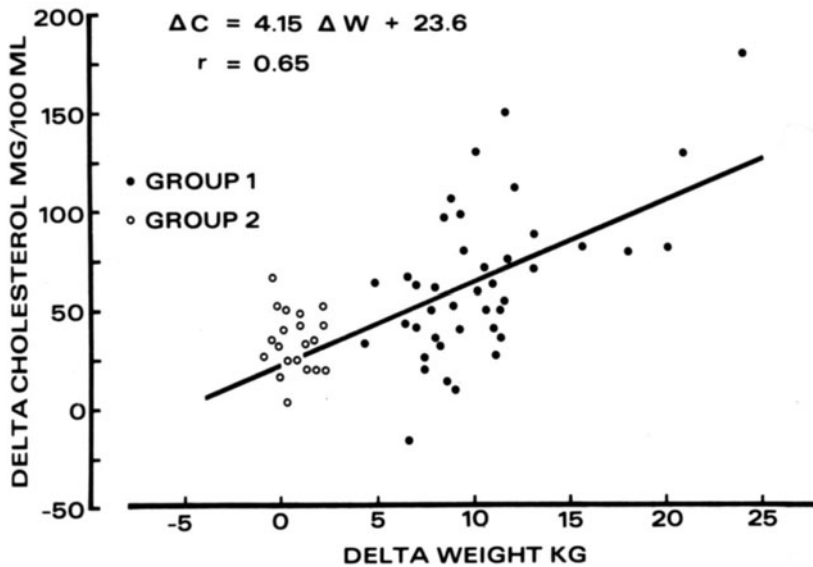


Figure 4: Change in serum cholesterol (Δc) plotted against change in body weight (Δw). Group 1 comprised 41 overweight hypercholesterolemic men who lost weight and Group 2 comprised 20 similar men whose weight did not change significantly during follow-up.

This study has brought home to us the almost insuperable difficulties of conducting intervention studies of this nature in coronary disease. Apart from the difficulties in the diet itself alterations in lifestyle which follow personal reappraisal introduce other variables which may have a greater influence than the dietary variables. It is neither practical nor ethical to advise men with coronary disease to continue to smoke, to ignore overweight and to remain physically inactive. Smoking, obesity and physical inactivity deserve brief comment.

The dramatic reduction in cigarette smoking in the whole cohort made the numbers of heavy smokers so few that the probable harmful effects were not apparent. In Gothenburg Wilhelmsen and his colleagues (6) have confirmed the expectation that continued smoking is unfavourable to survival in men with coronary disease.

The potential importance of weight change was not fully appreciated at the time this work was begun. There

is now good evidence that weight loss, particularly when combined with change in dietary lipids, has important favourable effects on both serum cholesterol and blood pressure (7,8,9). This is shown in Figure 4 which illustrates the relationship between weight loss and change in serum cholesterol in men from this study who were both obese and hypercholesterolemic (10). The fall in serum cholesterol in those who lost more than 5 kg in weight during follow-up was slightly more than double that of men whose weight was steady during follow-up. Both the P and F groups lost approximately 5 kg from infarct to the steady state during follow-up; the whole cohort had become lean by modern standards. This may account for the apparent lack of influence of body weight on survival. It may also explain in part the similar survival of the two dietary groups.

Recreational physical activity was shown to have a protective effect against heart disease in a recently reported prospective study of British civil servants (11). The favourable effect of physical activity observed in the present study is consistent with this. It was present after allowing for all other variables. This interesting finding merits further study.

Despite the imperfections of the present study the results are not very different from those reported by Leren (12) and by Dayton et al (13). The Leren study comprised 412 Norwegian men of similar age to our men and was done at a time when a control group could be maintained more easily on a normal western diet. There were a number of important differences in other features of the trial which cannot be discussed here. The dietary group lost 2.5 kg in weight and 52 mg/100 ml in serum cholesterol compared with 0.5 kg in weight and 11 mg/100 ml in serum cholesterol for the controls. Total 5 year cardiovascular mortality was 38 in the 206 dieted subjects and 52 in the controls. Sudden death accounted for 27 deaths in each group. Myocardial infarction caused 10 deaths in the diet group and 23 in the control group ($p < 0.03$). The annual death rate from heart and cerebrovascular disease was 3.7% in the diet group and 5.0% in the control group. Thus the possible saving by diet was 1.3 lives per 100 per year. In the Los Angeles study of Dayton et al (12) the men were much older and there was no improvement in overall five year survival in the diet group. It must be concluded that the lipid hypothesis has gained little support from secondary intervention studies.

Finally it is important to emphasize the remarkably good survival of unselected patients who have had myocardial infarction. The annual overall attrition rate of 3 to 4% per annum found here was similar to that in the Norwegian study (11), in the Coronary Drug Project (13) and in the Gothenburg study (6). How much this good survival is due merely to the natural history of the disease and how much is due to multiple changes in lifestyle after infarction is an open question.

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